Scientific computing/ MTH-373/573/2026
Homework

1. Compute QR factorization of

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 0 & 26 \\ 12 & 0 & -7 \\ 0 & 4 & 4 \\ 0 & -3 & -3 \end{bmatrix}$$

   using

   (a) Gram-Schimdt orthogonalization,

   (b) Givens rotations,

   (c) Householder reflections.

Assignment Problem (Graded)

1. Write codes using C/C++ to get QR factorization of a given matrix

   (a) using Gram-Schimdt orthogonalization,

   (b) Given's rotation,

   (c) Householder reflection.